Title 0:00–0:15. Joint work with Ken Brown.

Introduction

0:15–1:00 / 1 Outline Preview the cube, conditions, geometry, completion, and bounds.

1:00–2:30 / 2 Pauli strings X(a), Z(b) commute or anticommute. Stabilizers give the syndrome. Distance concerns nontrivial logical errors.

2:30–4:00 / 3 CSS codes X(a), $a \in C_2$, and Z(b), $b \in C_1^{\perp}$. Logical basis from S and K. Z syndrome Se^{T}.

4:00–5:15 / 4 Clifford, non-Clifford Gates on basis states. Clifford alone not universal. Add T or CCZ.

5:15–6:15 / 5 Transversal, distillation Fault stays on one qubit. Eastin–Knill. Distillation needs a transversal T or CCZ code.

6:15–7:30 / 6 Different codes, different gates Steane: Clifford. 15 qubits: T. Cube: CCZ. State the sign-layer question.

Cube example *7:30–7:40: outline, current theme highlighted.*

7:40–8:45 / 7 Points of $\mathbb{F}_2^3$ Reveal three coordinate faces, then the constant. Their span is RM$(1, 3)$.

8:45–9:30 / 8 Cube encoding S is all ones. K is the coordinates. Each logical state is a complement pair.

9:30–10:30 / 9 Cube distances An edge is a Z logical: d_Z=2. Nonconstant affine codewords give d_X=4.

10:30–11:30 / 10 Signed weights Integer signed weight, mod 8. Every codeword in a logical coset needs the same phase.

11:30–13:00 / 11 Cube signs Put inverse gates on parity p. Distance to p is 4 except for the label 111 pair.

13:00–14:45 / 12 Face, edge, vertex Signed weights: face 0, edge 0, vertex -1. Only the cubic term survives.

14:45–15:30 / 13 Row congruences Coefficients 1, -2, 4 give mod 8, 4, 2. Only the logical triple is odd.

Gate conditions *15:30–15:40: outline, current theme highlighted.*

15:40–16:15 / 14 Historical context Distillation, row constraints, parent spaces, CSS diagonal gates. Keep brief.

16:15–17:30 / 15 Matrix versus space Matrix: distinct rows. Space: all triples, including repeats, so all vectors even.

17:30–19:00 / 16 Nine CH conditions Logical, mixed, stabilizer families. K3 odd. Parities allow a Clifford correction.

Geometric construction *19:00–19:10: outline, current theme highlighted.*

19:10–20:30 / 17 Q48 result 48 physical, 3 logical. d_X=16, d_Z=3. 26 T and 22 inverse, no correction.

20:30–21:30 / 18 Fix the stabilizers Projective columns. 8-divisible span. Find K. Solve signs. Divisibility is a choice.

21:30–22:30 / 19 Projective syndromes Nonzero columns detect one Z. Distinct columns detect two. This proves only d_Z.

22:30–23:30 / 20 48-point support 63 minus 15 gives 48. Remove $w = 00$. Three blocks of 16. Six coordinate rows.

23:30–25:00 / 21 Weights 24 and 32 Nonzero u coefficient: 3 times 8. Pure w coefficient: 2 times 16. Rank six.

25:00–26:00 / 22 Geometric family Recover 48 at $p = 4, q = 2$. Other counts are lower bounds. Star is a fixed-support census.

Column picture and completion *26:00–26:10: outline, current theme highlighted.*

26:10–27:30 / 23 Column indicators Occupied columns as a Boolean function. Cube: y. Q48: $w_1 + w_2 + w_1 w_2$.

27:30–29:00 / 24 Moments and monomials Overlap parity is a monomial moment. One exceptional cubic. State the coset briefly.

29:00–30:30 / 25 Logical labels Three label bits per point. Product forces $x = \kappa$. S-only constraints already hold.

30:30–32:30 / 26 Logical completion One-row space, compatible pairs, linear third row. Three-site witness makes d_Z=3.

32:30–34:00 / 27 $T/T^{\dagger}$ signs Binary gamma in mod-4 equations. Show the polynomial, count $6 + 6 + 10$ inverse gates.

Bounds and questions *34:00–34:10: outline, current theme highlighted.*

34:10–36:00 / 28 Reed–Muller supports Positive-degree tests vanish at 0. Correct weight parity, then use RM duality.

36:00–37:30 / 29 Exclusion through 38 Rank at most 8, support reductions, complete enumeration, no logical completion.

37:30–39:30 / 30 Length comparison Our 38 floor. WJS 39–46. 47 quasi-transversal. Sign failure is code-specific.

39:30–41:00 / 31 Takeaways and questions Geometry, labels, signs. Ask about higher distance and sign feasibility. Leave QR up.

41:00: finish and invite questions.